



Graph Autoencoders for Credit-Card Fraud Detection

Unsupervised learning on graphs and heterogeneous
graph auto-encoders — a critical review

Group 6



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1 Introduction

Why credit-card fraud detection is hard

2 Background

Graph-learning foundations for the model

3 Methodology

How the HGAE works, step by step

4 Experiments

Dataset, protocol, and metrics

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What the numbers reveal — read critically

6 Conclusion

Takeaways and future directions

The paper under review — and its data

Case study

Heterogeneous Graph Auto-Encoder for Credit Card Fraud Detection

Sudipta Majumder, Moirangthem Tiken Singh, Rabinder Kumar Prasad, et al. (7 authors)

International Journal of Computer Applications (IJCA), vol. 32, no. 2, pp. 124–138, June 2025

Contextualised via [snowball research](#): HGT, GAE/VGAE, GEM, MGAE, IrGAE

Dataset (public)

Kaggle [kartik2112/fraud-detection](#)

Simulated card transactions,
Jan 2019 – Dec 2020

1,000 customers • 800 merchants
1,852,394 transactions • **0.52% fraud**

Key takeaway

Same public data $\Rightarrow$ every claim in the paper is [checkable](#) — and we checked.

SECTION 1

Introduction

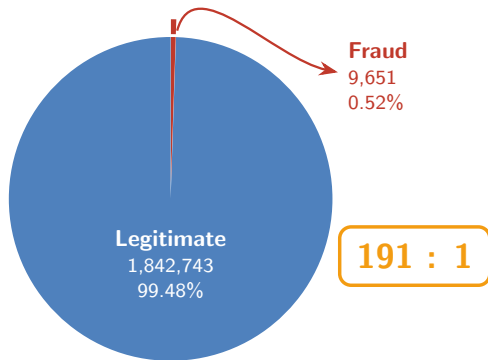
Why detecting credit-card fraud is hard — and why a label-free approach is needed.

Fraud is rare — and that is exactly the problem

- Digital payments exploded → sophisticated fraud
- Supervised ML is the default defence. . .
- . . . but fraud labels are **scarce, delayed, costly**
- Adversaries adapt → yesterday's labels age fast

Key takeaway

Learn what **normal** looks like instead of chasing known fraud.



Problem: extreme imbalance meets relational data

- **191 : 1** imbalance breaks supervised training
- Resampling (SMOTE) → **overfitting** / lost structure
- Transactions are **relational**, not i.i.d. rows

Research question

Can we detect fraud *without fraud labels* by learning the **topology** and **attributes** of legitimate behaviour?

Why existing methods fall short

Supervised / tabular

- Need many labels → infeasible
- Treat each transaction as **isolated**
- Ignore shared cards / merchants / bursts

What we actually need

- Works with **few / no labels**
- Captures **relational** structure
- Stays **scalable** & **interpretable**

Key takeaway

Topology is precisely the signal row-wise classifiers discard.

Objectives & contributions

1. **Survey**: from message-passing GNNs to self-supervised GAEs
2. **Dissect**: the Heterogeneous Graph Auto-Encoder (HGAE)
3. **Evaluate**: architecture, training, reported results — critically
4. **Expose**: formulation & reproducibility gaps (*our contribution*)
5. **Propose**: research agenda for robust detection (*our contribution*)

SECTION 2

Background

The graph-learning foundations needed to understand the model.

HGAE stands on two intellectual lines



Heterogeneous attention

HGT, GEM — type/relation-aware

Reconstruction self-supervision

GAE, VGAE — no labels

Background: message passing, attention, and its limits

- **Message passing**: a node refines itself by aggregating neighbours
- **GAT**: learns *how much* each neighbour matters
- **Permutation equivariance**: the right inductive bias for networks

Caveat

Power bounded by the **1-WL** test; deep stacks cause **oversmoothing**. (*Both return in our critique.*)

The core idea: reconstruction error as an anomaly score

- Encoder compresses graph $\rightarrow$ latent Z
- Decoder reconstructs the input
- Train on **normal** data only
- Anomalies $\rightarrow$ **high reconstruction error**

One-class framing

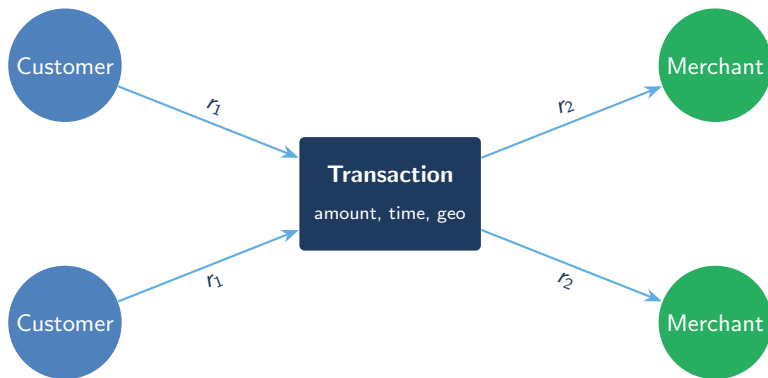
Never having modelled fraud, the decoder rebuilds it *poorly* — the error betrays it.

SECTION 3

Methodology

How the Heterogeneous Graph Auto-Encoder turns transactions into anomaly scores.

Step 1 — Lift transactions into a heterogeneous graph



Three node types, typed edges r_1, r_2

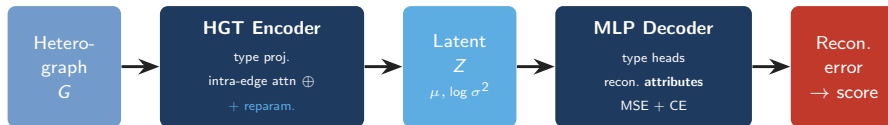
IDs → connectivity only (not features)
is_fraud excluded from training

Step 2 — Encode attributes into a shared space

- Continuous (amount, time, geo) → **standardised**
- Categorical (gender, category) → **one-hot**
- Missing → per-type **mean / mode**
- Concatenate ⇒ layer-0 input h^0

Three types ⇒ different dimensions ⇒ reconciled by **type-specific linear projections**.

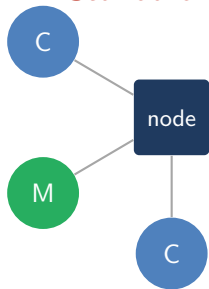
Step 3 — The HGAE pipeline at a glance



- **Encoder** (HGT): type projections + intra-edge attention + reparameterization
- **Latent Z** : a smooth, probabilistic embedding space
- **Decoder** reconstructs **attributes**, *not* adjacency $\rightarrow$ error = score

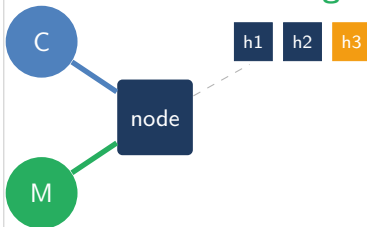
Key novelty — intra-edge attention respects entity types

Standard GAT



softmax across all neighbours
Customer competes with Merchant

HGAE intra-edge



softmax across the H heads of one edge
selects the informative subspace

$$f_{\text{Attent}} = \text{Softmax}(\|_{k \in [1, H]} \text{Att}^k(v_t, e_r, d_t)) \text{ — normalised over heads, not neighbours}$$

Step 4 — A probabilistic latent space

- Reparameterization at the final encoder layer
- Each embedding = a **Gaussian** (mean, log-var)
- Noise $\rightarrow$ **smooth** space; sampling stays **differentiable**

$$h = \text{mean}(h) + \epsilon \cdot \exp\left(\frac{1}{2} \log(h)\right), \quad \epsilon \sim \mathcal{N}(0, 1)$$

Caveat

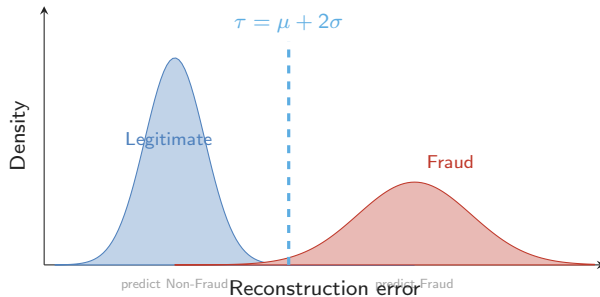
Loss has **no KL term** $\Rightarrow$ strictly a noise-injected AE, not a true VAE.

Step 5 — Decode, then threshold the error

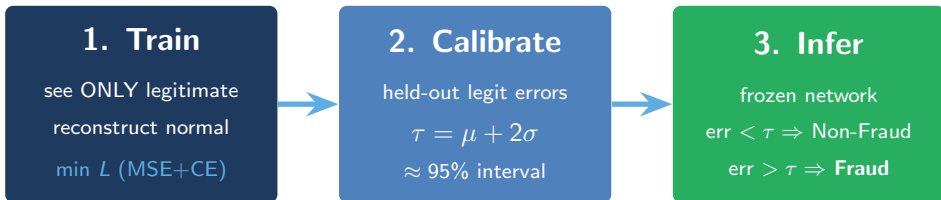
- Type-specific MLP decoder
- err below $\tau \Rightarrow$ **Non-Fraud**
- err above $\tau \Rightarrow$ **Fraud**

$$\tau = \mu + 2\sigma$$

$\approx$ 95% interval of normal error



Step 6 — Three-phase training, no fraud labels in the loss



$$L = \sum_{\forall N} \sum_{\forall t} \text{LOSS}(d_t, d'_t)$$

MSE for continuous attributes + **CE** for categorical attributes

SECTION 4

Experiments

The dataset, protocol, and metrics used to put the model to the test.

Dataset — a realistic, highly imbalanced log

Property	Value
Genuine transactions	1,842,743
Fraudulent	9,651 ($\approx 0.52\%$ — $\sim 191:1$)
Held-out test set	553,574 genuine / 2,145 fraud
Node types	Customer, Merchant, Transaction
Key fields	cc_num, merchant_id, amount, time, geo
Label usage	calibration & evaluation only

Setup & why AUC-PR is the metric that matters

Setup

- Train on **legitimate-only** split
- Calibrate τ on held-out legit
- Baselines: **GraphSAGE**, **FI-GRL**

Metrics

- **AUC-PR** — key under imbalance
- **F1** — precision/recall at τ
- ROC-AUC — **optimistic** when negatives dominate

Caveat

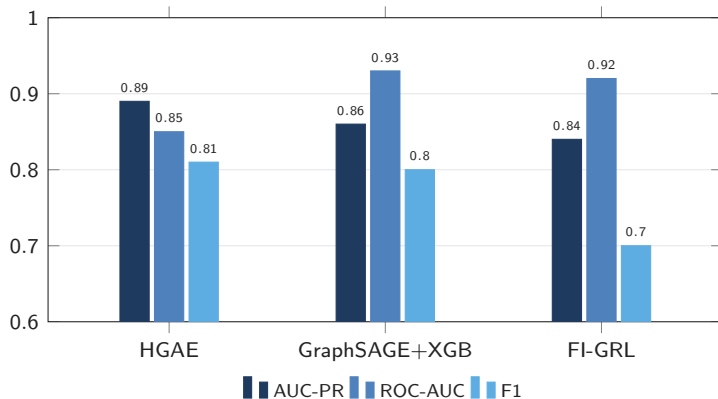
Reported encoder depth = **124 layers** — implausible under oversmoothing; blocks reproduction.

SECTION 5

Results

What the numbers reveal — and a critical reading of their limits.

Results — a precision–recall vs. ROC trade-off



HGAE leads on AUC-PR & F1 — but its ROC-AUC is the lowest.

Comparison — superiority is nuanced, not uniform

Method	AUC-PR	F1	ROC-AUC
HGAE	0.89	0.81	0.85
GraphSAGE + XGBoost	0.86	0.80	0.93
FI-GRL	0.84	0.70	0.92

Verdict

Wins where it matters under imbalance (**AUC-PR**, **F1**), but **loses on ROC-AUC** — a trade-off the original paper never reconciles.

Discussion — five formulation concerns

1. **Implausible depth**: 124 layers vs. oversmoothing $\rightarrow$ not reproducible
2. **Attribute-only decoding**: structure enters the score only *indirectly*
3. **Incomplete VAE**: no KL term $\rightarrow$ weak “variational” claim
4. **Overstated superiority**: lowest ROC-AUC, never reconciled
5. **Ambiguous threshold**: $\mu + 2\sigma$ vs. F1-sweep $\rightarrow$ possible leakage

Key takeaway

#2 is the deepest: it undercuts the paper's own “topology is what others miss”.

Limitations inherited from the GNN family

- **Scalability**: $O(nE)$ full-graph infeasible
- **Static-graph fallacy**: blind to velocity attacks
- **Heterophily**: fraud camouflages among legit
- **No explainability**: opaque score
- **Oversmoothing / -squashing** with depth
- **Latent collapse**: no contrastive term

These map back to the **challenges** raised at the start — the loop closes.

SECTION 6

Conclusion

Key takeaways and a research agenda for robust, label-free fraud detection.

Future work — every fix targets a named limitation

- **Masked & contrastive GAEs** (MGAE, IrGAE) → collapse + attribute-only
- **Architectural simplification** (linear prop., orthogonal init) → scale + depth
- **Temporal heterogeneous GNNs** → velocity / evolving fraud
- **Adaptive thresholds** → drop the global Gaussian assumption
- **Explainable attribution** → per-attribute loss, attention maps

- GAEs reframe fraud as **one-class anomaly detection** — beating label scarcity
- HGAE fuses **type-specific attention** + unsupervised reconstruction
- But superiority is **nuanced**; formulation & GNN limits remain
- Forward: **masked/contrastive, temporal, simplified, explainable**

Key takeaway

Learn what *normal* looks like on the graph and let fraud betray itself — but verify every claim honestly.

Thank you — Questions?

References

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